

Indian Institute of Technology Jammu
Department of Electrical Engineering
Mid-Semester Examination
Basic Electrical and Electronics Engineering EE-101



Q1. When connected to a 120-V (rms), 60-Hz power line, a load absorbs 4 kW at a lagging power factor of 0.8. Find the value of capacitance necessary to raise the pf to 0.95. [5Marks]

Q2. If, for the given figure.1, load resistance (R_L) and Thevenin equivalent resistance (R_{th}) are both reduced by 30% from the earlier value, then power delivered to the load is increased or decreased by what percentage of the maximum power delivered? Draw the graph between power delivered to load (P_L) vs. R_L for both cases. [5Marks]

Q3. In the given figure 2, switch S1 is closed at $t = 0$. When the capacitor voltage reaches 98% of its final value, switch S1 is opened, and switch S2 is closed at the same time and remains closed for 40 msec. [5Marks]

- A. Determine the voltage across the capacitor after 40 msec of closing S2.
- B. What is the value of the voltage across the capacitor after closing switch S1 at $t = 4\tau$?
- C. Find the charge stored in the capacitor just after i) opening S1 and ii) after 40 msec of closing S2.
- D. Plot the voltage across the capacitor vs. time, marking proper voltage levels and time points clearly.

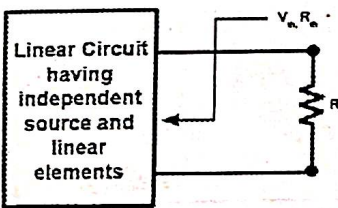


Figure 1.

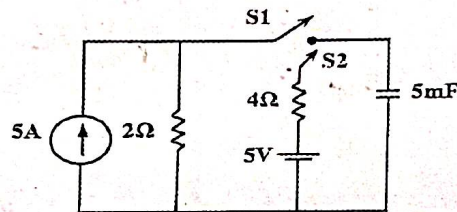


Figure 2.

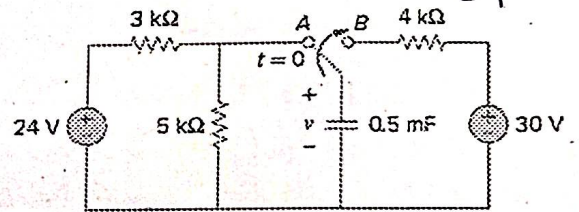


Figure 3.

Q4. The switch shown in the figure 3, has remained in position A for an extended period. At $t = 0$, the switch moves to B. Determine $v(t)$ for $t > 0$ and calculate its value at $t = 1$ and 4 s. [5Marks]

Q5. A dc circuit shown in figure 4, has a voltage V , a current source I , and several resistors. A particular resistor R dissipates a power of 4 W when V alone is active. The same resistor dissipates a power of 9W when I alone is active. Calculate the power dissipated by R when both sources are active. [5Marks]

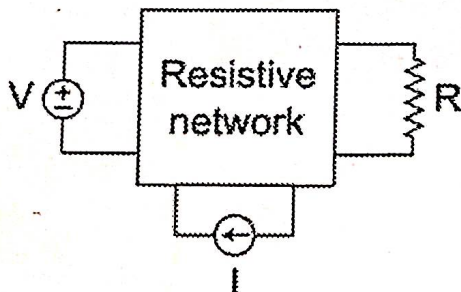


Figure 4.

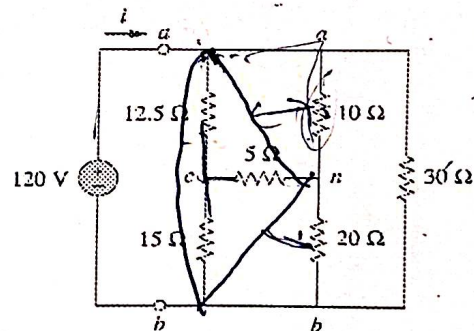


Figure 5.

$$(I_A + I_R)^2 R = V$$

$$\frac{4}{7} \frac{V_{th}^2 R_L}{R_L + R_{th}} = \frac{V_{th}^2}{4 R_{th}}$$

$$I_A^2 R = 4$$

$$I_B^2 R = 9$$

$$I_A = \frac{2}{\sqrt{R}}$$

$$I_B = \frac{3}{\sqrt{R}}$$